



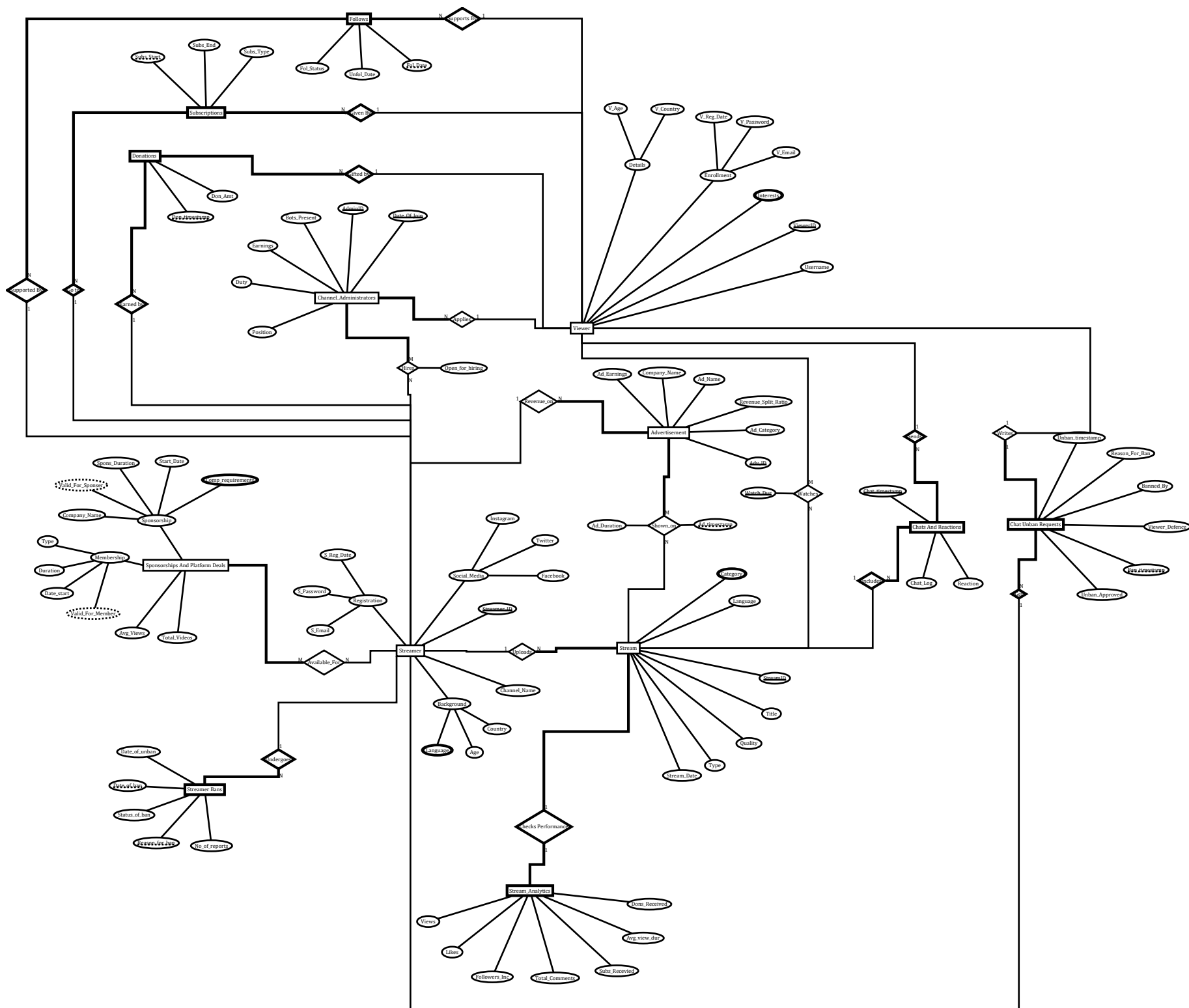
Database Project - StreamWave

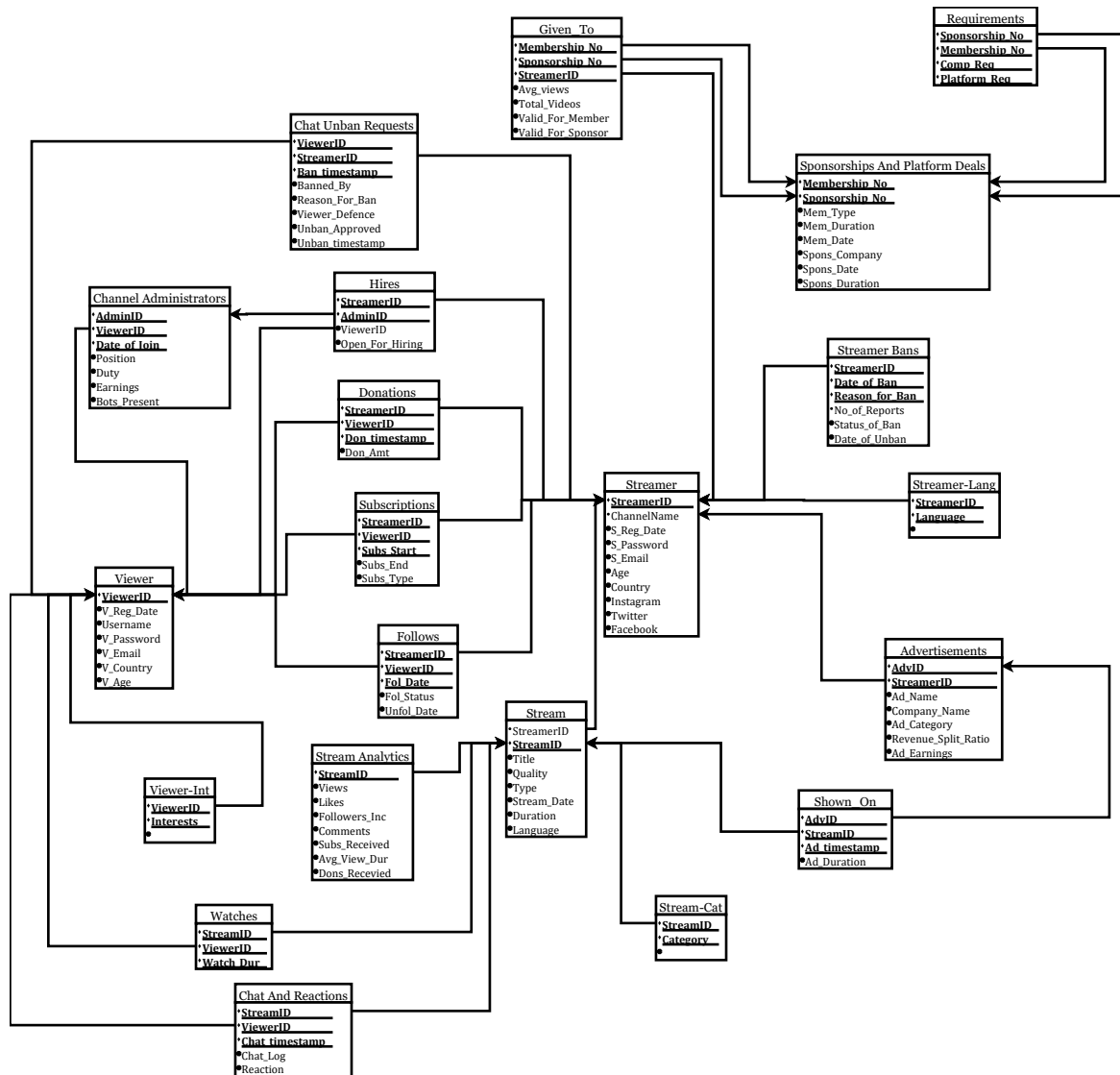
(Based on an Online Live Streaming Platform)

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Note: In the ER Diagram the bold branches represent mandatory relationship





Minimal Functional Dependency Sets and Normal Forms in StreamWave's Database:-

Streamer Relation:

- StreamerID $\rightarrow$ ChannelName
- StreamerID $\rightarrow$ S_Reg_Date
- StreamerID $\rightarrow$ S_Password
- StreamerID $\rightarrow$ S_Email
- StreamerID $\rightarrow$ Age
- StreamerID $\rightarrow$ Country
- StreamerID $\rightarrow$ Instagram
- StreamerID $\rightarrow$ Twitter
- StreamerID $\rightarrow$ Facebook
- $(\text{StreamerID})^+ = (\text{StreamID}, \text{ChannelName}, \text{S_Reg_Date}, \text{S_Email}, \text{Age}, \text{Country}, \text{Instagram}, \text{Twitter}, \text{Facebook})$
- Primary Key: StreamerID as its closure includes all attributes of Streamer Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Streamer, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Streamer-Lang Relation:

- $\{\text{StreamerID}, \text{Language}\} \rightarrow \text{Language}$
- $\{\text{StreamerID}, \text{Language}\} \rightarrow \text{StreamerID}$
- $(\text{StreamerID}, \text{Language})^+ = (\text{StreamerID}, \text{Language})$
- Composite Key: (StreamID, Language) as its closure includes all attributes of Streamer-Lang Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Streamer-Lang, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form). Also, as all the attributes of the relation form the composite key, the relation will be in BCNF.

Viewer Relation:

- ViewerID $\rightarrow$ V_Reg_Date
- ViewerID $\rightarrow$ V_Password
- ViewerID $\rightarrow$ V_Country
- ViewerID $\rightarrow$ V_Age
- ViewerID $\rightarrow$ Username

- $(\text{ViewerID})^+ = (\text{ViewerID}, \text{V_Reg_Date}, \text{V_Password}, \text{V_Country}, \text{V_Age}, \text{Username})$
- Primary Key: ViewerID as its closure includes all attributes of Viewer Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Viewer, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Viewer-Int Relation:

- $\{\text{ViewerID}, \text{Interests}\} \rightarrow \text{Interests}$
- $\{\text{ViewerID}, \text{Interests}\} \rightarrow \text{ViewerID}$
- $(\text{ViewerID}, \text{Interests})^+ = (\text{ViewerID}, \text{Interests})$
- Composite Key: (ViewerID, Interests) as its closure includes all attributes of Viewer-Int Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Viewer-Int, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form). Also, as all the attributes of the relation form the composite key, the relation will be in BCNF.

Stream Relation:

- StreamID $\rightarrow$ Title
- StreamID $\rightarrow$ Quality
- StreamID $\rightarrow$ Type
- StreamID $\rightarrow$ Stream_Date
- StreamID $\rightarrow$ Duration
- StreamID $\rightarrow$ Language
- StreamID $\rightarrow$ StreamerID
- $(\text{StreamID})^+ = (\text{StreamID}, \text{Title}, \text{Quality}, \text{Type}, \text{Stream_Date}, \text{Duration}, \text{Language}, \text{StreamerID})$
- Primary Key: StreamID as its closure includes all attributes of Stream Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Stream, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Stream-Cat Relation:

- $\{\text{StreamID}, \text{Category}\} \rightarrow \text{Category}$
- $\{\text{StreamID}, \text{Category}\} \rightarrow \text{Stream}$
- $(\text{StreamID}, \text{Category})^+ = (\text{StreamID}, \text{Category})$
- Composite Key: (StreamID, Category) as its closure includes all attributes of Stream-Cat Relation.

- For every Functional Dependency $A \rightarrow B$ that holds on the relation Stream-Cat, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form). Also, as all the attributes of the relation form the composite key, the relation will be in BCNF.

Stream Analytics Relation:

- StreamID $\rightarrow$ Views
- StreamID $\rightarrow$ Likes
- StreamID $\rightarrow$ Comments
- StreamID $\rightarrow$ Followers_Inc
- StreamID $\rightarrow$ Subs_Received
- StreamID $\rightarrow$ Dons_Received
- StreamID $\rightarrow$ Avg_View_Dur
- $(\text{StreamID})^+ = (\text{StreamID}, \text{Views}, \text{Likes}, \text{Comments}, \text{Followers_Inc}, \text{Subs_Received}, \text{Dons_Received}, \text{Avg_View_Dur})$
- Primary Key: StreamID as its closure includes all attributes of Stream Analytics Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Stream Analytics, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Streamer Bans Relation:

- $\{\text{StreamerID}, \text{Date_of_Ban}, \text{Reason_for_Ban}\} \rightarrow \text{No_of_Reports}$
- $\{\text{StreamerID}, \text{Date_of_Ban}, \text{Reason_for_Ban}\} \rightarrow \text{Status_of_Ban}$
- $\{\text{StreamerID}, \text{Date_of_Ban}, \text{Reason_for_Ban}\} \rightarrow \text{Date_of_Unban}$
- $(\text{StreamerID}, \text{Date_of_Ban}, \text{Reason_for_Ban})^+ = (\text{StreamerID}, \text{Date_of_Ban}, \text{Reason_for_Ban}, \text{No_of_Reports}, \text{Status_of_Ban}, \text{Date_of_Unban})$
- Composite Key: $(\text{StreamerID}, \text{Date_of_Ban}, \text{Reason_for_Ban})$ as its closure includes all attributes of Streamer Bans Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Streamer Bans, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Chat And Reactions:

- $\{\text{StreamerID}, \text{ViewerID}, \text{Chat_timestamp}\} \rightarrow \text{Chat_Log}$
- $\{\text{StreamerID}, \text{ViewerID}, \text{Chat_timestamp}\} \rightarrow \text{Reaction}$
- $(\text{StreamerID}, \text{ViewerID}, \text{Chat_timestamp})^+ = (\text{StreamerID}, \text{ViewerID}, \text{Chat_timestamp}, \text{Chat_Log}, \text{Reaction})$

- Composite Key: (StreamerID, ViewerID, Chat_timestamp) as its closure includes all attributes of Chat And Reactions Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Chat and Reactions, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Chat Unban Requests:

- {StreamerID, ViewerID, Ban_timestamp} -> Banned_by
- {StreamerID, ViewerID, Ban_timestamp} -> Reason_For_Ban
- {StreamerID, ViewerID, Ban_timestamp} -> Viewer_Defence
- {StreamerID, ViewerID, Ban_timestamp} -> Unban_Approved
- {StreamerID, ViewerID, Ban_timestamp} -> Unban_timestamp
- (StreamerID, ViewerID, Ban_timestamp)⁺ = (StreamerID, ViewerID, Ban_timestamp, Banned_by, Reason_For_Ban, Viewer_Defence, Unban_Approved, Unban_timestamp)
- Composite Key: (StreamerID, ViewerID, Ban_timestamp) as its closure includes all attributes of Chat Unban Requests Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Chat Unban Requests, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Channel Administrators:

- {AdminID, ViewerID, Date_of_Join} -> Position
- {AdminID, ViewerID, Date_of_Join} -> Duty
- {AdminID, ViewerID, Date_of_Join} -> Earnings
- {AdminID, ViewerID, Date_of_Join} -> Bots_Present
- (AdminID, ViewerID, Date_of_Join)⁺ = (AdminID, ViewerID, Date_of_Join, Position, Duty, Earnings, Bots_Present)
- Composite Key: (AdminID, ViewerID, Date_of_Join) as its closure includes all attributes of Channel Administrators Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Channel Administrators, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Hires:

- {StreamerID, AdminID} -> Open_For_Hiring

- $(\text{StreamerID}, \text{AdminID})^+ = (\text{StreamerID}, \text{AdminID}, \text{Open_For_Hiring})$
- Composite Key: $(\text{StreamerID}, \text{AdminID})$ as its closure includes all attributes of Hires Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Hires, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Advertisements:

- $\{\text{AdvID}, \text{StreamerID}\} \rightarrow \text{Ad_Name}$
- $\{\text{AdvID}, \text{StreamerID}\} \rightarrow \text{Company_Name}$
- $\{\text{AdvID}, \text{StreamerID}\} \rightarrow \text{Ad_Category}$
- $\{\text{AdvID}, \text{StreamerID}\} \rightarrow \text{Revenue_Split}$
- $\{\text{AdvID}, \text{StreamerID}\} \rightarrow \text{Ad_Earnings}$
- $(\text{AdvID}, \text{StreamerID})^+ = (\text{AdvID}, \text{StreamerID}, \text{Ad_Name}, \text{Company_Name}, \text{Ad_Category}, \text{Revenue_Split}, \text{Ad_Earnings})$
- Composite Key: $(\text{AdvID}, \text{StreamerID})$ as its closure includes all attributes of Advertisements Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Advertisements, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Shown On:

- $\{\text{AdvID}, \text{StreamID}, \text{Ad_Timestamp}\} \rightarrow \text{Ad_Duration}$
- $(\text{AdvID}, \text{StreamID}, \text{Ad_Timestamp})^+ = (\text{AdvID}, \text{StreamID}, \text{Ad_Timestamp}, \text{Ad_Duration})$
- Composite Key: $(\text{AdvID}, \text{StreamID}, \text{Ad_Timestamp})$ as its closure includes all attributes of Shown_On Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Shown_On, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Sponsorships And Platform Deals:

- $\{\text{Membership_No}, \text{Sponsorship_No}\} \rightarrow \text{Mem_Type}$
- $\{\text{Membership_No}, \text{Sponsorship_No}\} \rightarrow \text{Mem_Duration}$
- $\{\text{Membership_No}, \text{Sponsorship_No}\} \rightarrow \text{Mem_Date}$
- $\{\text{Membership_No}, \text{Sponsorship_No}\} \rightarrow \text{Spons_Company}$
- $\{\text{Membership_No}, \text{Sponsorship_No}\} \rightarrow \text{Spons_Date}$
- $\{\text{Membership_No}, \text{Sponsorship_No}\} \rightarrow \text{Spons_Duration}$

- $(\text{MembershipNo}, \text{Sponsorship_No})^+ = (\text{MembershipNo}, \text{Sponsorship_No}, \text{Mem_Type}, \text{Mem_Duration}, \text{Mem_Date}, \text{Spons_Company}, \text{Spons_Date}, \text{Spons_Duration})$
- Composite Key: $(\text{Membership_No}, \text{Sponsorship_No})$ as its closure includes all attributes of Sponsorships and Platform Deals Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Sponsorship and Platform Deals, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Given_To:

- $\{\text{Membership_No}, \text{Sponsorship_No}, \text{StreamerID}\} \rightarrow \text{Avg_views}$
- $\{\text{Membership_No}, \text{Sponsorship_No}, \text{StreamerID}\} \rightarrow \text{Total_Videos}$
- $\{\text{Membership_No}, \text{Sponsorship_No}, \text{StreamerID}\} \rightarrow \text{Valid_For_Member}$
- $\{\text{Membership_No}, \text{Sponsorship_No}, \text{StreamerID}\} \rightarrow \text{Valid_For_Sponsor}$
- $(\text{Membership_No}, \text{Sponsorship_No}, \text{StreamerID})^+ = (\text{Membership_No}, \text{Sponsorship_No}, \text{StreamerID}, \text{Avg_Views}, \text{Total_Videos}, \text{Valid_For_Member}, \text{Valid_For_Sponsor})$
- Composite Key: $(\text{Membership_No}, \text{Sponsorship_No}, \text{StreamerID})$ as its closure includes all attributes of Given_To Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Given_To, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Requirements:

- $\{\text{Sponsorship_No}, \text{Membership_No}, \text{Comp_Req}, \text{Platform_Req}\} \rightarrow \text{Sponsorship_No}$
- $\{\text{Sponsorship_No}, \text{Membership_No}, \text{Comp_Req}, \text{Platform_Req}\} \rightarrow \text{Membership_No}$
- $\{\text{Sponsorship_No}, \text{Membership_No}, \text{Comp_Req}, \text{Platform_Req}\} \rightarrow \text{Comp_Req}$
- $\{\text{Sponsorship_No}, \text{Membership_No}, \text{Comp_Req}, \text{Platform_Req}\} \rightarrow \text{Platform_Req}$
- $(\text{Sponsorship_No}, \text{Membership_No}, \text{Comp_Req}, \text{Platform_Req})^+ = (\text{Sponsorship_No}, \text{Membership_No}, \text{Comp_Req}, \text{Platform_Req})$
- Composite Key:
 $(\text{Sponsorship_No}, \text{Membership_No}, \text{Comp_Req}, \text{Platform_Req})$ as its closure includes all attributes of Requirements Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Requirements, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form). Also, as all the attributes of the relation form the composite key, the relation will be in BCNF.

Donations:

- $\{\text{StreamerID}, \text{ViewerID}, \text{Don_timestamp}\} \rightarrow \text{Don_Amt}$
- $(\text{StreamerID}, \text{ViewerID}, \text{Don_timestamp})^+ = (\text{StreamerID}, \text{ViewerID}, \text{Don_timestamp}, \text{Don_Amt})$
- Composite Key: $(\text{StreamerID}, \text{ViewerID}, \text{Don_timestamp})$ as its closure includes all attributes of Donations Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Donations, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Subscriptions:

- $\{\text{StreamerID}, \text{ViewerID}, \text{Subs_Start}\} \rightarrow \text{Subs_End}$
- $\{\text{StreamerID}, \text{ViewerID}, \text{Subs_Start}\} \rightarrow \text{Subs_Type}$
- $(\text{StreamerID}, \text{ViewerID}, \text{Subs_Start})^+ = (\text{StreamerID}, \text{ViewerID}, \text{Subs_Start}, \text{Subs_End}, \text{Subs_Type})$
- Composite Key: $(\text{StreamerID}, \text{ViewerID}, \text{Subs_Start})$ as its closure includes all attributes of Subscriptions Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Subscriptions, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Follows:

- $\{\text{StreamerID}, \text{ViewerID}, \text{Fol_Date}\} \rightarrow \text{Fol_Status}$
- $\{\text{StreamerID}, \text{ViewerID}, \text{Fol_Date}\} \rightarrow \text{Unfol_Date}$
- $(\text{StreamerID}, \text{ViewerID}, \text{Fol_Date})^+ = (\text{StreamerID}, \text{ViewerID}, \text{Fol_Date}, \text{Fol_Status}, \text{Unfol_Date})$
- Composite Key: $(\text{StreamerID}, \text{ViewerID}, \text{Fol_Date})$ as its closure includes all attributes of Follows Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Follows, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form).

Watches:

- $\{\text{StreamID}, \text{ViewerID}, \text{Watch_Dur}\} \rightarrow \text{StreamID}$
- $\{\text{StreamID}, \text{ViewerID}, \text{Watch_Dur}\} \rightarrow \text{ViewerID}$
- $\{\text{StreamID}, \text{ViewerID}, \text{Watch_Dur}\} \rightarrow \text{Watch_Dur}$
- $(\text{StreamID}, \text{ViewerID}, \text{Watch_Dur})^+ = (\text{StreamID}, \text{ViewerID}, \text{Watch_Dur})$

- Composite Key: (StreamID, ViewerID, Watch_Dur) as its closure includes all attributes of Watches Relation.
- For every Functional Dependency $A \rightarrow B$ that holds on the relation Watches, A is its super key, hence the relation is in BCNF (Boyce-Codd Normal Form). Also, as all the attributes of the relation form the composite key, the relation will be in BCNF.